

Ziliang Shen

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Shanghai, China

OBJECTIVE

I'm currently pursuing a Ph.D. in Statistics at Shanghai University of Finance and Economics. I am passionate about statistical and machine learning theories, particularly in field of **high-dimensional statistical learning, quantile regression, reinforcement learning in finance**, and **distributed statistical inference**.

With a calm demeanor and a passion for tech innovation, I thrive on learning and applying new insights in my work.

EDUCATION

- **Shanghai University of Finance and Economics** 2021.09 - Now
PhD in Statistics Shanghai, China
- **Nanchang University** 2015.09 - 2019.06
Bachelor in Mathematics and Applied Mathematics (Financial Mathematics) Nanchang, China

EXPERIENCE

- **Turing Fund Managment** [🌐] 2025.03 - Now
Reinforcement Learning Quantitative Research Shanghai, China
 - Developed a high-frequency stock trading execution model achieving a measurable reduction in execution slippage within a simulated Level 3 data environment.
 - Implemented DQN and PPO algorithms with hierarchical reinforcement learning, enhancing sample efficiency and overall strategy stability.
 - Conducted analysis on real market data, identifying delayed-reward patterns and key factors influencing model robustness.

PROJECTS

- **Shanghai Electric Solar Cell Target Detection Project** 2024.05 - 2024.06
Tools: *Faster R-CNN, PyTorch, Labeling Tools*
 - Led university team in collaboration with Shanghai Hongpu Information Technology, coordinating on-site work
 - Directed data labeling and iteratively optimized Faster R-CNN-based detection model to improve accuracy and efficiency
- **Shanghai GDP Nowcasting Project** 2024.06 - 2024.10
Tools: *R, Python, Dynamic Factor Model, MF-BAVART, LSTM*
 - Developed multiple nowcasting models to address missing data and mixed-frequency issues, enabling real-time GDP growth rate estimation

TEACHING EXPERIENCE

- **Teaching Assistant** 2022.09 - Present
 - Undergraduate Courses: Advanced Mathematical Statistics I & II, Linear Model, Distributed Computation.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

- [C.1] Caixing Wang, Ziliang Shen. (2024). **Distributed high-dimensional quantile regression: estimation efficiency and support recovery**. In *International Conference on Machine Learning (ICML)*, Spotlight. [Code]
- [S.1] Caixing Wang, Ziliang Shen, Shaoli Wang, Xingdong Feng. **Estimation and Inference on Distributed High-Dimensional Quantile Regression: Double-Smoothing and Debiasing**. Manuscript submitted to *Journal of Machine Learning Research (JMLR)*.
- [S.2] Hanteng Ma, Ziliang Shen, Xingdong Feng, Xin Liu. **Sparsity learning via structured functional factor augmentation**. Manuscript submitted to *Statistica Sinica*.
- [S.3] Ziliang Shen, Caixing Wang, Shaoli Wang, Yibo Yan. **High-dimensional differentially private quantile regression: distributed estimation and statistical inference**. Manuscript submitted to *IEEE Transactions on Information Theory*.

HONORS AND AWARDS

- **First-class Undergraduate Scholarship**
Nanchang University
- **First-class Graduate Scholarship**
Shanghai University of Finance and Economics
- **Excellent Student Award & Excellent Member of the Communist Youth League**
Shanghai University of Finance and Economics
- **National Scholarship for Doctoral Students**
Ministry of Education, China

ACTIVITIES

- **17th China R Conference & 2024 X Intelligence Conference**
Renmin University of China
◦ Participated in *New Statistical Methods I* session and engaged in academic exchange

Jul 2024
- **Shanghai Doctoral Forum on Statistics**
East China Normal University
◦ Awarded Second Prize for research presentation

Nov 2024
- **3rd National Conference on Statistics and Data Science**
Zhejiang University
◦ Presented research poster and exchanged ideas with peers

Jul 2025

PROFESSIONAL MEMBERSHIPS

- **IEEE student membership** 2025.07 - Present

REFERENCES

1. **Jianhua Chen**
Counselor, Department of Mathematics and Computer Science
Nanchang University
Email: jianhuachen@ncu.edu.cn
Relationship: Undergraduate counsellor
2. **Mei Zhuang**
Deputy Secretary of the Party Committee
School of Statistics and Data Science, Shanghai University of Finance and Economics
Email: zhuangmm@mail.shufe.edu.cn
Relationship: Doctoral counsellor
3. **Shaoli Wang**
Associate Professor
School of Statistics and Data Science, Shanghai University of Finance and Economics
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Relationship: advisor